

22

Cost-Volume-Profit

Learning Objectives

1

Explain variable, fixed, and mixed costs and the relevant range.

2

Apply the high-low method to determine the components of mixed costs.

3

Prepare a CVP income statement to determine contribution margin.

4

Compute the break-even point using three approaches.

5

Determine the sales required to earn target net income and determine margin of safety.

6

Use CVP analysis to respond to changes in the business environment.

Cost Behavior Analysis is the study of how specific costs respond to changes in the level of business activity.

- ◆ Some costs change; others remain the same.
- ◆ Helps management plan operations and decide between alternative courses of action. Applies to all types of businesses and entities.
- ◆ Starting point is measuring key business activities. Changes in the level or volume of activity should be correlated with changes in costs.
- ◆ Activity level selected is called activity or volume index.

Activity index:

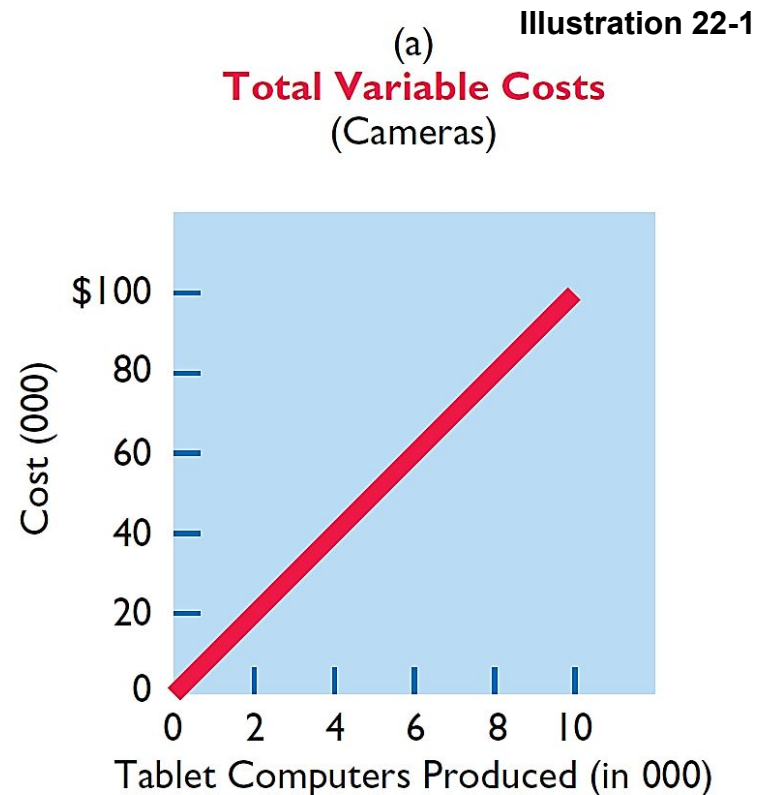
- ◆ Identifies the activity that causes changes in the behavior of costs. Allows costs to be classified as variable, fixed, or mixed.

Variable Costs

- ◆ Costs that vary in total directly and proportionately with changes in the activity level.
 - ▶ **Example:** If the activity level increases 10 percent, total variable costs increase 10 percent.
 - ▶ **Example:** If the activity level decreases by 25 percent, total variable costs decrease by 25 percent.
- ◆ Variable costs **remain the same** per unit at every level of activity.

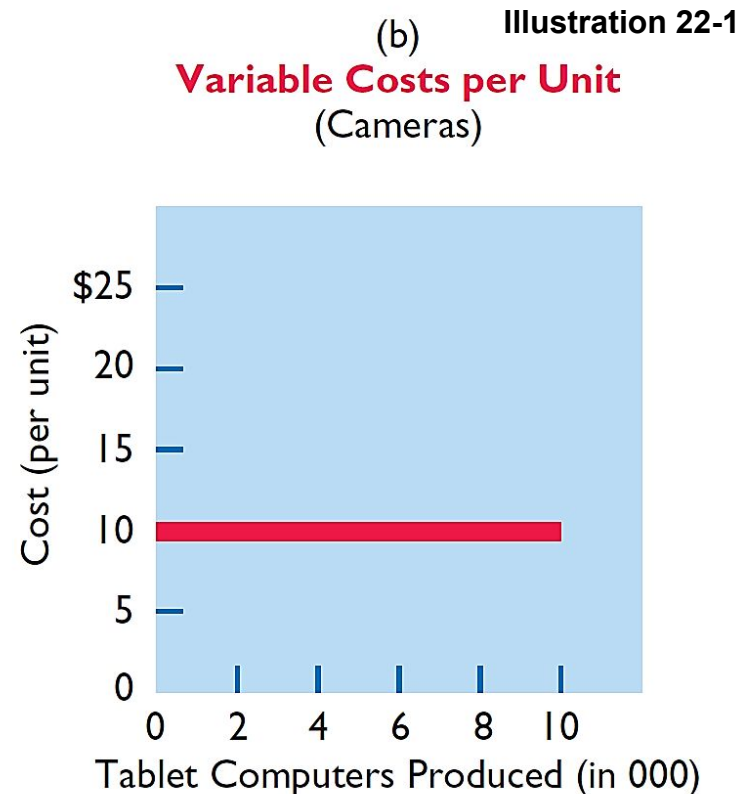
Variable Costs

Illustration: Damon Company manufactures tablet computers that contain a \$10 camera. The activity index is the number of tablets produced. As Damon manufactures each tablet, the total cost of the cameras used increases by \$10. **As part (a)** of Illustration 22-1 shows, total cost of the cameras will be \$20,000 if Damon produces 2,000 tablets, and \$100,000 when it produces 10,000 tablets. We also can see that a variable cost remains the same per unit as the level of activity changes.



Variable Costs

Illustration: Damon Company manufactures tablet computers that contain a \$10 camera. The activity index is the number of tablets produced. As Damon manufactures each tablet, the total cost of the cameras used increases by \$10. **As part (b)** of Illustration 22-1 shows, the unit cost of \$10 for the camera is the same whether Damon produces 2,000 or 10,000 tablets.



Variable Costs

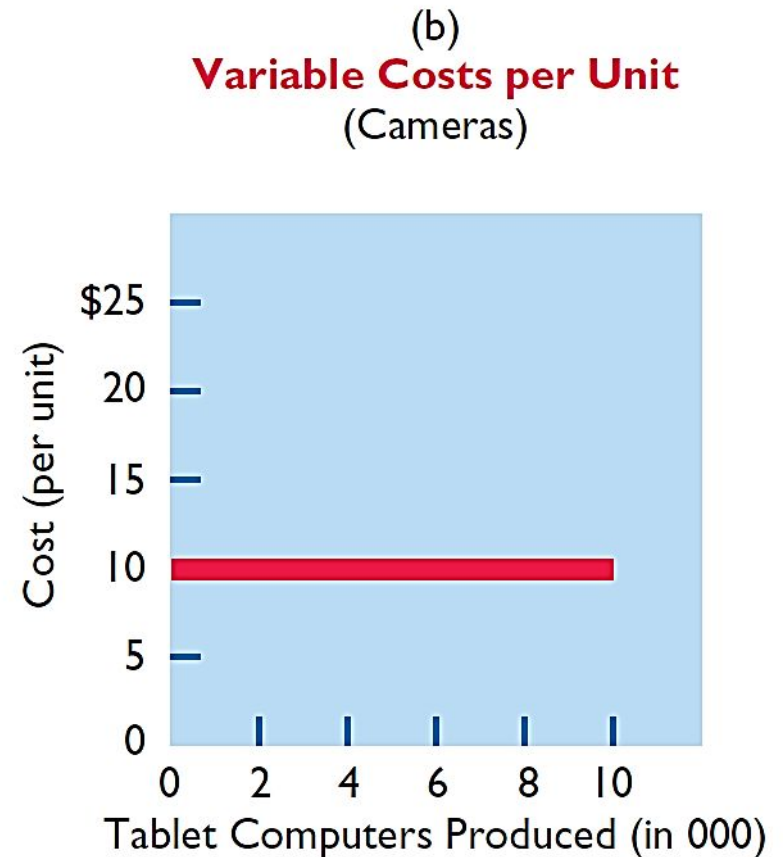
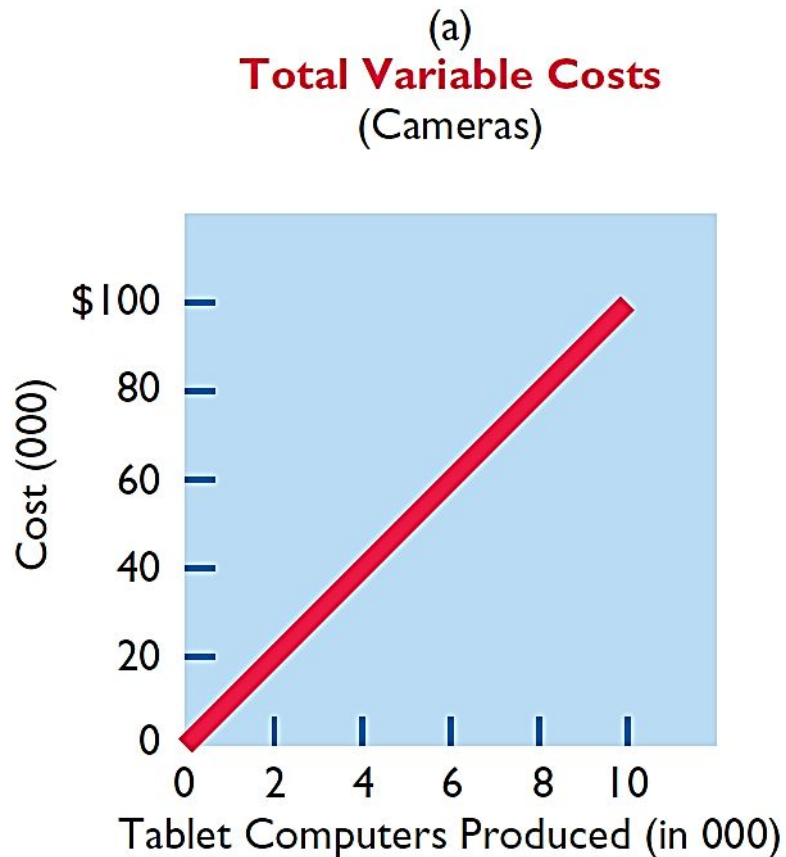


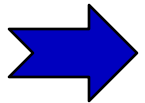
Illustration 22-1
Behavior of total and
unit variable costs

Variable Costs

Question

Variable costs are costs that:

- a. Vary in total directly and proportionately with changes in the activity level.
- b. Remain the same per unit at every activity level.
- c. Neither of the above.
- d. Both (a) and (b) above.

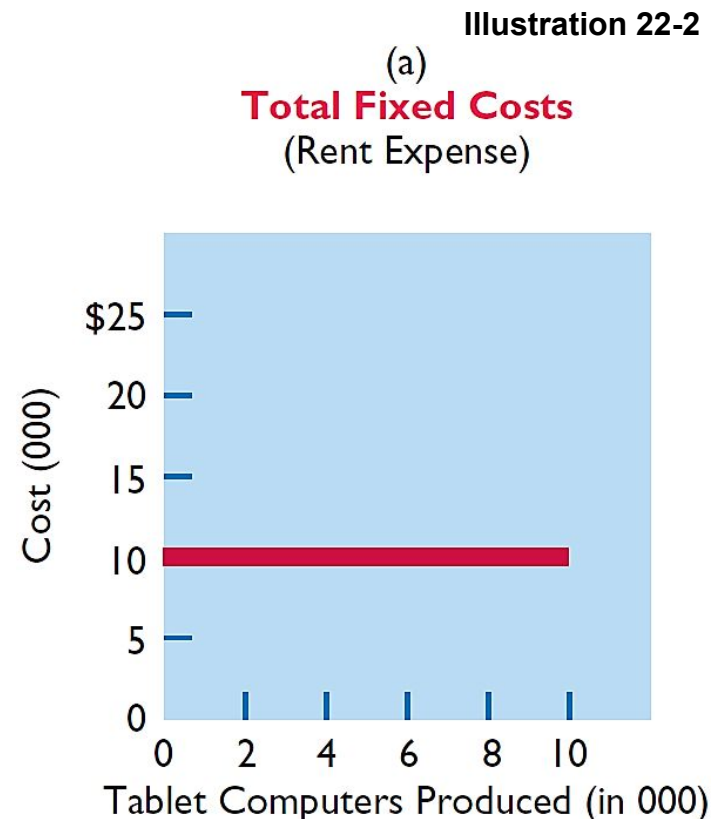


Fixed Costs

- ◆ **Costs that remain the same** in total regardless of changes in the activity level within a relevant range.
- ◆ **Fixed cost per unit cost varies inversely** with activity:
As volume increases, unit cost declines, and vice versa
- ◆ **Examples:**
 - ▶ Property taxes
 - ▶ Insurance
 - ▶ Rent
 - ▶ Depreciation on buildings and equipment

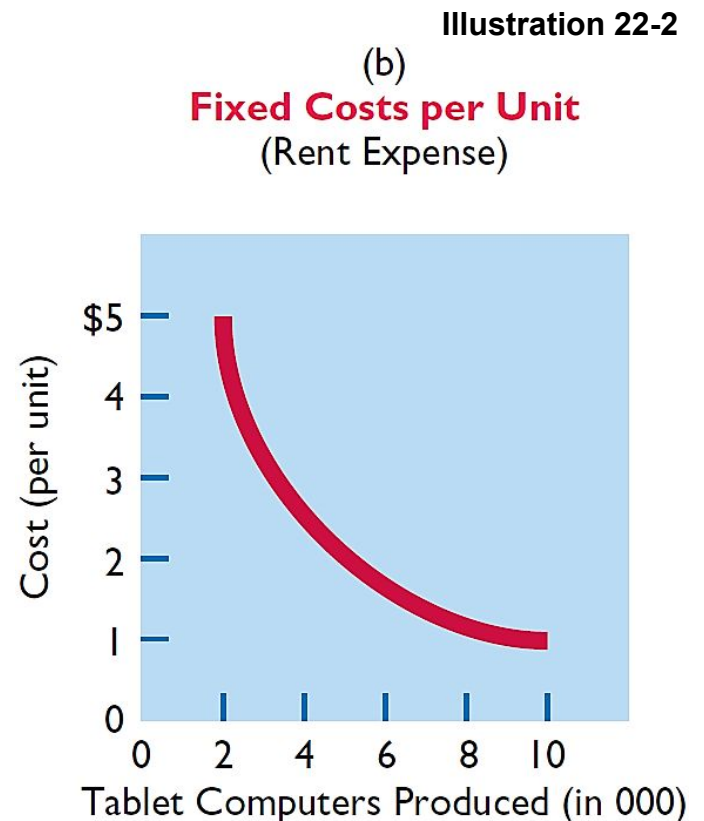
Fixed Costs

Illustration: Damon Company leases its productive facilities at a cost of \$10,000 per month. Total fixed costs of the facilities will remain constant at every level of activity, as part (a) of Illustration 22-2 shows.



Fixed Costs

Illustration: Damon Company leases its productive facilities at a cost of \$10,000 per month. Total fixed costs of the facilities will remain constant at every level of activity. But, **on a per unit basis, the cost of rent will decline as activity increases**, as part (b) of Illustration 22-2 shows. At 2,000 units, the unit cost per tablet computer is \$5 ($\$10,000 \div 2,000$). When Damon produces 10,000 tablets, the unit cost of the rent is only \$1 per tablet ($\$10,000 \div 10,000$).



Fixed Costs

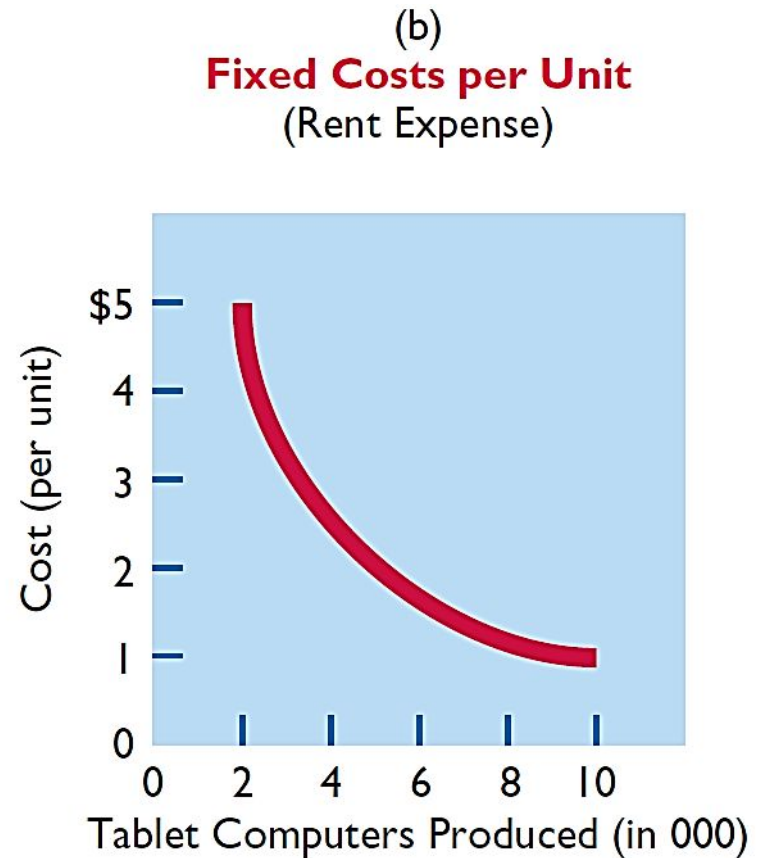
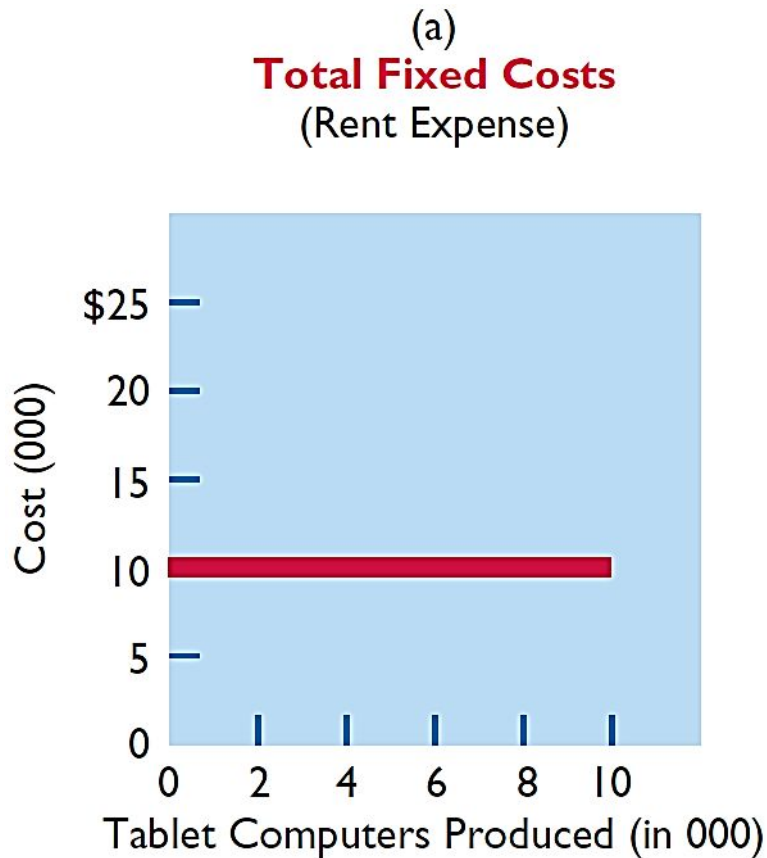


Illustration 22-2
Behavior of total and
unit fixed costs

Relevant Range

- ◆ Throughout the range of possible levels of activity, a straight-line relationship usually does not exist for either variable costs or fixed costs.
- ◆ Relationship between variable costs and changes in activity level is often **curvilinear**.
- ◆ For **fixed costs**, the **relationship is also nonlinear** – some fixed costs will not change over the entire range of activities, while other fixed costs may change.

Helpful Hint

Fixed costs that may be changeable include research, such as new product development, and management training programs.

Relevant Range

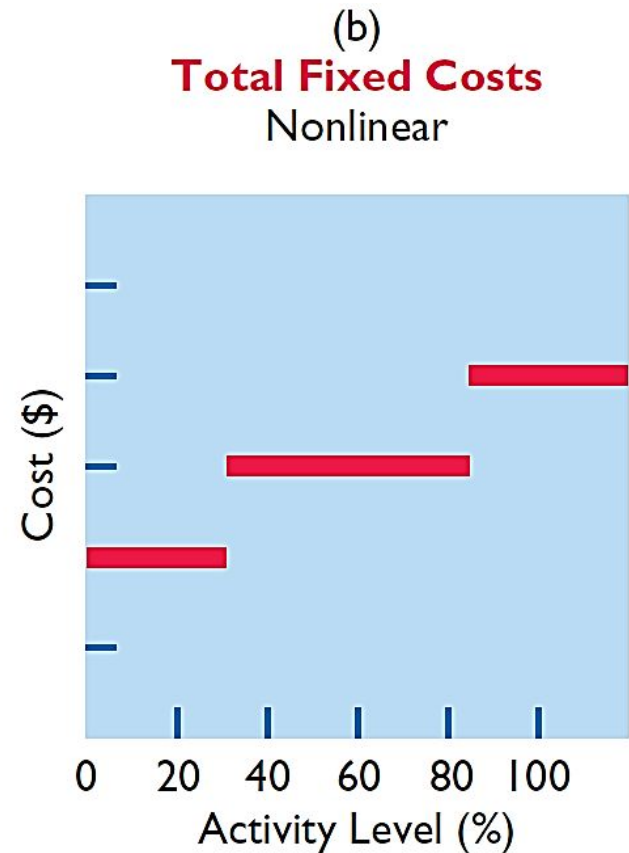
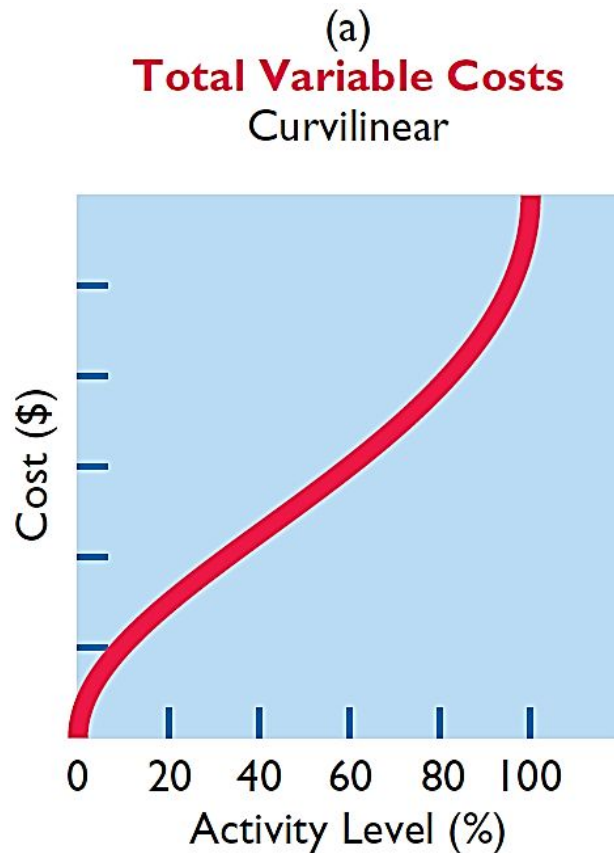
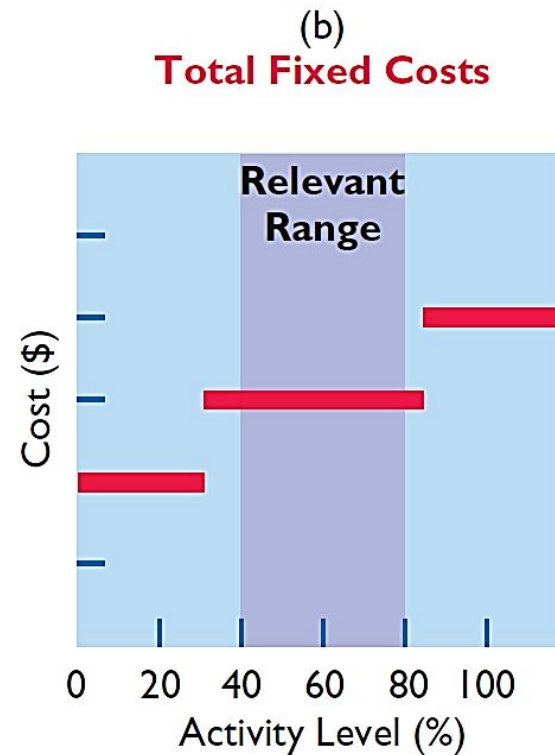
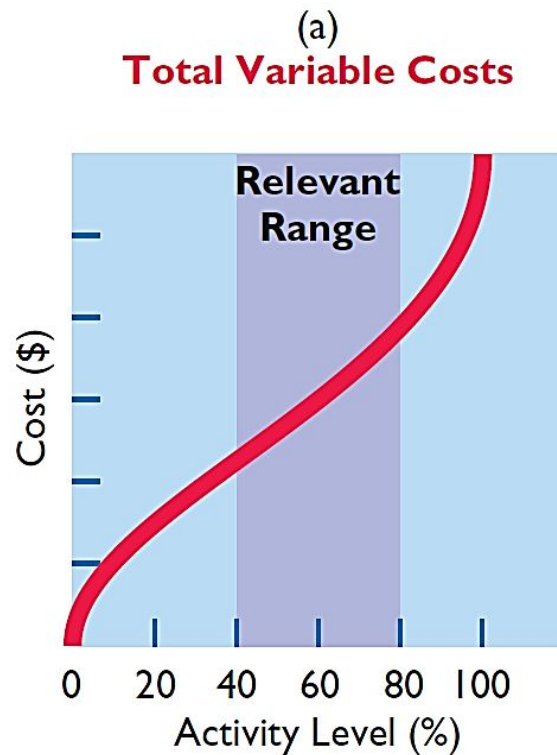


Illustration 22-3
Nonlinear behavior of
variable and fixed costs

Relevant Range

Range of activity over which a company expects to operate during a year.

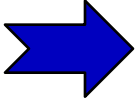
Illustration 22-4
Linear behavior within
relevant range



Relevant Range

Question

The relevant range is:

- a. The range of activity in which variable costs will be curvilinear.
- b. The range of activity in which fixed costs will be curvilinear.
-  c. The range over which the company expects to operate during a year.
- d. Usually from zero to 100% of operating capacity.

Mixed Costs

- ◆ Costs that have **both** a variable element **and** a fixed element.
- ◆ Change in total but not proportionately with changes in activity level.

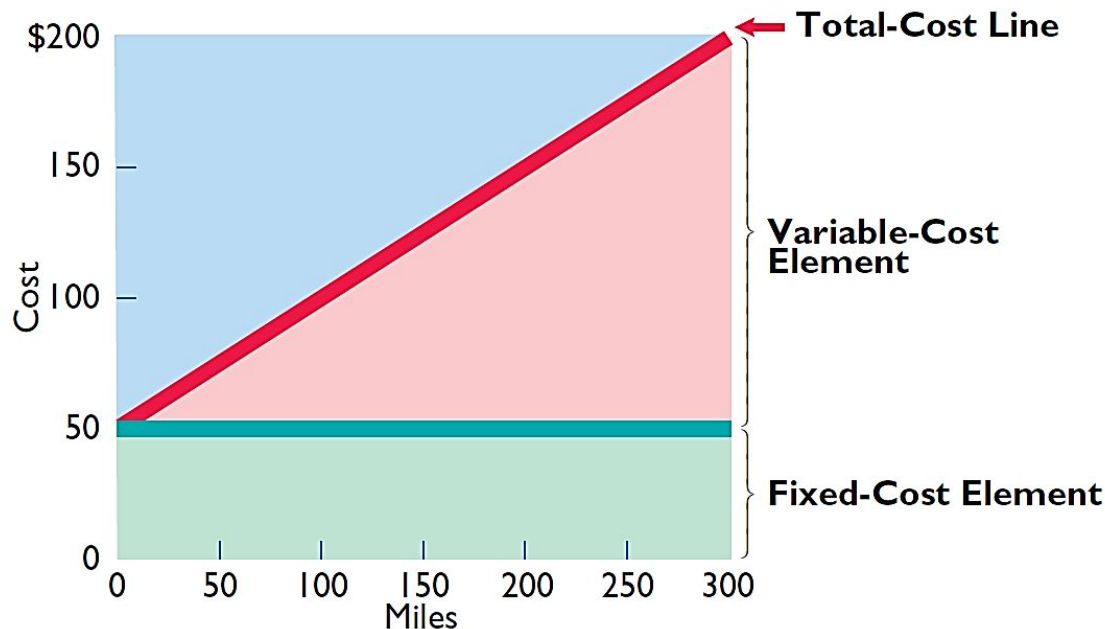


Illustration 22-5
Behavior of a mixed cost

DO IT!

1

Type of Costs

Helena Company, reports the following total costs at two levels of production.

Classify each cost as **variable**, **fixed**, or **mixed**.

	<u>10,000 Units</u>	<u>20,000 Units</u>
Variable →	Direct materials \$20,000	\$40,000
→	Maintenance 8,000	10,000
→	Direct labor 17,000	34,000
→	Indirect materials 1,000	2,000
Fixed →	Depreciation 4,000	4,000
→	Utilities 3,000	5,000
Mixed →	Rent 6,000	6,000

High-Low Method

- ◆ High-Low Method uses the total costs incurred at the high and the low levels of activity to classify mixed costs into fixed and variable components.
- ◆ The difference in costs between the high and low levels represents variable costs, since only variable-cost element can change as activity levels change.

STEP 1: Determine variable cost per unit using the following formula:

$$\text{Change in Total Costs} \div \text{High minus Low Activity Level} = \text{Variable Cost per Unit}$$

High-Low Method

Illustration: Metro Transit Company has the following maintenance costs and mileage data for its fleet of buses over a 6-month period.

<u>Month</u>	<u>Miles Driven</u>	<u>Total Cost</u>	<u>Month</u>	<u>Miles Driven</u>	<u>Total Cost</u>
January	20,000	\$30,000	April	50,000	\$63,000
February	40,000	48,000	May	30,000	42,000
March	35,000	49,000	June	43,000	61,000

Change in Costs	(63,000 - 30,000)	\$33,000	=	\$1.10 cost per unit
High minus Low	(50,000 - 20,000)	<u>30,000</u>		

High-Low Method

STEP 2: Determine the **fixed cost** by subtracting the total variable cost at **either** the high or the low activity level from the total cost at that activity level.

Illustration 22-8
High-low method
computation of
fixed costs

	A	B	C	D	
1	METRO TRANSIT				
2	Activity Level				
3			High	Low	
4	Total cost		\$63,000	\$30,000	
5	Less:	Variable costs			
6		$50,000 \times \$1.10$	55,000		
7		$20,000 \times \$1.10$		22,000	
8	Total fixed costs		\$ 8,000	\$ 8,000	
9					
10					

High-Low Method

Maintenance costs are therefore \$8,000 per month of fixed costs plus \$1.10 per mile of variable costs. This is represented by the following formula:

$$\text{Maintenance costs} = \$8,000 + (\$1.10 \times \text{Miles driven})$$

Example: At 45,000 miles, estimated maintenance costs would be:

Fixed	\$ 8,000	
Variable	(\$1.10 x 45,000)	49,500
		\$57,500

High-Low Method

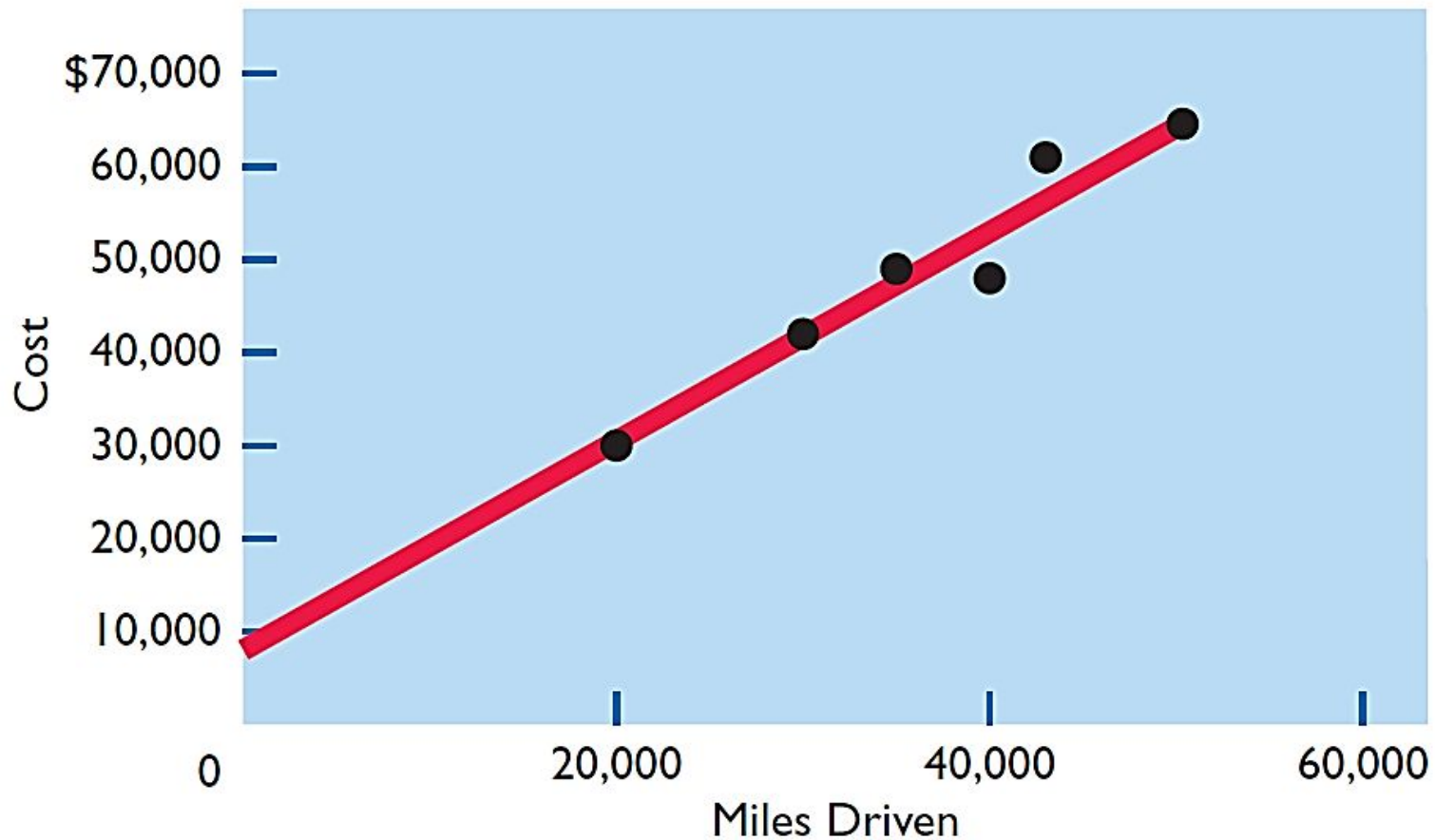
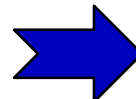


Illustration 22-9
Scatter plot for Metro
Transit Company

High-Low Method

Question

Mixed costs consist of a:

- 
- a. Variable cost element and a fixed cost element.
 - b. Fixed cost element and a controllable cost element.
 - c. Relevant cost element and a controllable cost element.
 - d. Variable cost element and a relevant cost element.

DO IT!**2**

High-Low Method

Byrnes Company accumulates the following data concerning a mixed cost, using units produced as the activity level.

	<u>Units Produced</u>	<u>Total Cost</u>
March	9,800	\$14,740
April	8,500	13,250
May	7,000	11,100
June	7,600	12,000
July	8,100	12,460

- (a) Compute the variable- and fixed-cost elements using the high-low method.
- (b) Estimate the total cost if the company produces 8,000 units.

DO IT!

2

High-Low Method

	<u>Units Produced</u>	<u>Total Cost</u>
March	9,800	\$14,740
April	8,500	13,250
May	7,000	11,100
June	7,600	12,000
July	8,100	12,460

- (a) Compute the variable and fixed cost elements using the high-low method.

Variable cost: $(\$14,740 - \$11,100) / (9,800 - 7,000) = \text{\$1.30 per unit}$

Fixed cost: $\$14,740 - \$12,740 (\$1.30 \times 9,800 \text{ units}) = \text{\$2,000}$

or $\$11,100 - \$9,100 (\$1.30 \times 7,000) = \text{\$2,000}$

DO IT!

2

High-Low Method

	<u>Units Produced</u>	<u>Total Cost</u>
March	9,800	\$14,740
April	8,500	13,250
May	7,000	11,100
June	7,600	12,000
July	8,100	12,460

(b) Estimate the total cost if the company produces 8,000 units.

Total cost (8,000 units): $\$2,000 + \$10,400 (\$1.30 \times 8,000) = \textbf{\$12,400}$

Cost-volume-profit (CVP) analysis is the study of the effects of changes in costs and volume on a company's profits.

- ◆ Important in profit planning.
- ◆ Critical factor in management decisions as
 - ▶ Setting selling prices,
 - ▶ Determining product mix, and
 - ▶ Maximizing use of production facilities.

Cost-Volume-Profit Analysis

Basic Components

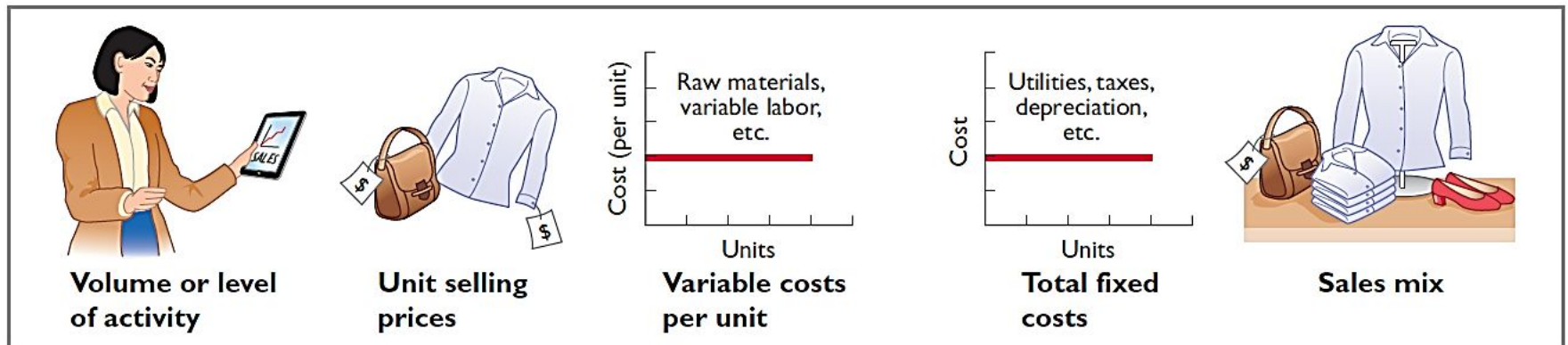


Illustration 22-10
Components of CVP analysis

Basic Components

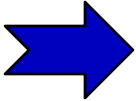
Assumptions

- ◆ Behavior of both costs and revenues is linear throughout the relevant range of the activity index.
- ◆ Costs can be classified accurately as either variable or fixed.
- ◆ Changes in activity are the only factors that affect costs.
- ◆ All units produced are sold.
- ◆ When more than one type of product is sold, the sales mix will remain constant.

Basic Components

Question

Which of the following is **not** involved in CVP analysis?

- a. Sales mix.
- b. Unit selling prices.
-  c. Fixed costs per unit.
- d. Volume or level of activity.

Cost-Volume-Profit Analysis

CVP Income Statement

- ◆ A statement for **internal use**.
- ◆ Classifies costs and expenses as fixed or variable.
- ◆ Reports **contribution margin** in the body of the statement.
 - ▶ **Contribution margin** – amount of revenue remaining after deducting variable costs.
- ◆ Reports the **same net income** as a traditional income statement.

CVP Income Statement

Illustration: Vargo Video Company produces a high-definition digital camcorder. Relevant data for the camcorders sold by this company in June 2014 are as follows.

Unit selling price of camcorder	\$500
Unit variable costs	\$300
Total monthly fixed costs	\$200,000
Units sold	1,600

Illustration 22-11

Assumed selling and cost data
for Vargo Video

CVP Income Statement

Illustration: The CVP income statement for Vargo Video therefore would be reported as follows.

VARGO VIDEO COMPANY	
CVP Income Statement	
For the Month Ended June 30, 2017	
	Total
Sales (1,600 camcorders)	\$ 800,000
Variable costs	480,000
Contribution margin	320,000
Fixed costs	200,000
Net income	\$120,000

Illustration 22-12

CVP Income Statement

UNIT CONTRIBUTION MARGIN

- ◆ Contribution margin is available to **cover fixed costs** and to **contribute to income**.
- ◆ Formula for **contribution margin per unit** and the computation for Vargo Video are:

Unit Selling Price	—	Unit Variable Costs	=	Unit Contribution Margin
\$500	—	\$300	=	\$200

Illustration 22-13

Formula for unit contribution margin

CVP Income Statement

UNIT CONTRIBUTION MARGIN

Vargo's CVP income statement assuming a zero net income.

VARGO VIDEO COMPANY CVP Income Statement For the Month Ended June 30, 2017		
	<u>Total</u>	<u>Per Unit</u>
Sales (1,000 camcorders)	\$ 500,000	\$ 500
Variable costs	<u>300,000</u>	<u>300</u>
Contribution margin	200,000	<u>\$200</u>
Fixed costs	<u>200,000</u>	
Net income	<u>\$ -0-</u>	

Illustration 22-14

CVP Income Statement

UNIT CONTRIBUTION MARGIN

Assume that Vargo sold one more camcorder, for a total of 1,001 camcorders sold.

Illustration 22-15

VARGO VIDEO COMPANY CVP Income Statement For the Month Ended June 30, 2017		
	<u>Total</u>	<u>Per Unit</u>
Sales (1,001 camcorders)	\$500,500	\$ 500
Variable costs	<u>300,300</u>	<u>300</u>
Contribution margin	200,200	<u>\$200</u>
Fixed costs	<u>200,000</u>	
Net income	<u>\$ 200</u>	

CVP Income Statement

Unit Selling Price	—	Unit Variable Costs	=	Unit Contribution Margin
\$500	—	\$300	=	\$200

Unit Contribution Margin	÷	Unit Selling Price	=	Contribution Margin Ratio
\$200	÷	\$500	=	40%

Illustration 22-17

Formula for contribution margin ratio

CVP Income Statement

CONTRIBUTION MARGIN RATIO

VARGO VIDEO COMPANY CVP Income Statement For the Month Ended June 30, 2017		
	<u>Total</u>	<u>Percent of Sales</u>
Sales (1,001 camcorders)	\$500,500	100%
Variable costs	<u>300,300</u>	<u>60</u>
Contribution margin	200,200	40%
Fixed costs	<u>200,000</u>	<u></u>
Net income	<u>\$ 200</u>	

Illustration 22-16

CVP income statement, with
net income and percent of sales data

CVP Income Statement

CONTRIBUTION MARGIN RATIO

Assume Vargo Video's current sales are \$500,000 and it wants to know the effect of a \$100,000 (200-unit) increase in sales.

VARGO VIDEO COMPANY CVP Income Statements For the Month Ended June 30, 2017						
	No Change			With Change		
	Total	Per Unit	Percent of Sales	Total	Per Unit	Percent of Sales
Sales	\$500,000	\$ 500	100%	\$600,000	\$ 500	100%
Variable costs	300,000	300	60	360,000	300	60
Contribution margin	200,000	\$200	40%	240,000	\$200	40%
Fixed costs	200,000			200,000		
Net income	\$ -0-			\$ 40,000		

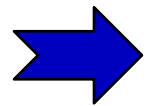
Illustration 22-18

CVP Income Statement

Question

Contribution margin:

- a. Is revenue remaining after deducting variable costs.
- b. May be expressed as contribution margin per unit.
- c. Is selling price less cost of goods sold.
- d. Both (a) and (b) above.



DO IT!

3

CVP Income Statement

Ampco Industries produces and sells a cell phone-operated thermostat. Information regarding the costs and sales of thermostats during September 2017 are provided below.

Unit selling price of thermostat \$85

Unit variable costs \$32

Total monthly fixed costs \$190,000

Units sold 4,000

Prepare a CVP income statement for Ampco Industries for the month of September. Provide per unit values and total values.

DO IT!

3

CVP Income Statement

Prepare a CVP income statement for Ampco Industries for the month of September. Provide per unit values and total values.

Solution

AMPCO INDUSTRIES
CVP Income Statement
For the Month Ended September 30, 2017

	<u>Total</u>	<u>Per Unit</u>
Sales		
Variable costs		
Contribution margin		—
Fixed costs		
Net income	—	

Break-Even Analysis

- ◆ Process of finding the **break-even point** level of activity at **which total revenues equal total costs** (both fixed and variable).
- ◆ Can be computed or derived
 - ▶ from a **mathematical equation**,
 - ▶ by using **contribution margin**, or
 - ▶ from a **cost-volume profit (CVP) graph**.
- ◆ Expressed either in **sales units** or in **sales dollars**.

Compute the break-even point using three approaches.

Mathematical equation

Required Sales – variable costs-Fixed Cost=Net Income

Contribution Margin Technique

(CM = total revenues – variable costs)

Fixed Cost/ Contribution Margin per Unit=Net Break-Even Point in Units

Fixed Cost/ Contribution Margin Ratio=Net Break-Even Point in Dollars

Unit Selling Price	–	Unit Variable Costs	=	Unit Contribution Margin
\$500	–	\$300	=	\$200

Unit Contribution Margin	÷	Unit Selling Price	=	Contribution Margin Ratio
\$200	÷	\$500	=	40%

Target Net Income and Margin of Safety

Mathematical equation

Required Sales – variable costs – Fixed Cost = Target Net Income

Contribution Margin Technique

$\left(\begin{array}{c} \text{Fixed Costs +} \\ \text{Target Net Income} \end{array} \right)$	$\div$	Contribution Margin per Unit	=	Required Sales in Units
(\$200,000 + \$120,000)	$\div$	\$200	=	1,600 units

$\left(\begin{array}{c} \text{Fixed Costs +} \\ \text{Target Net Income} \end{array} \right)$	$\div$	Contribution Margin Ratio	=	Required Sales in Dollars
(\$200,000 + \$120,000)	$\div$	40%	=	\$800,000

Actual (Expected) Sales	—	Break-Even Sales	=	Margin of Safety in Dollars
\$750,000	—	\$500,000	=	\$250,000

Actual (Expected) Sales	—	Break-Even Sales	=	Margin of Safety in Dollars
\$750,000	—	\$500,000	=	\$250,000

P22-2B All Frute Company bottles and distributes Frute Ade, a fruit drink. The beverage is sold for 50 cents per 16-ounce bottle to retailers, who charge customers 70 cents per bottle. For the year 2014, management estimates the following revenues and costs.

Sales	\$2,500,000	Selling expenses—variable	\$ 80,000
Direct materials	360,000	Selling expenses—fixed	250,000
Direct labor	450,000	Administrative expenses—	
Manufacturing overhead—		variable	40,000
variable	270,000	Administrative expenses—	
Manufacturing overhead—		fixed	150,000
fixed	380,000		

Instructions

- Prepare a CVP income statement for 2014 based on management's estimates. (Show column for total amounts only.)
- Compute the break-even point in (1) units and (2) dollars.
- Compute the contribution margin ratio and the margin of safety ratio.
- Determine the sales dollars required to earn net income of \$624,000.

Mathematical Equation

Break-even occurs where total sales equal variable costs plus fixed costs; i.e., net income is zero

Computation
of
break-even
point in units.

Required Sales	–	Variable Costs	–	Fixed Costs	=	Net Income
\$500Q	–	\$300Q	–	\$200,000	=	\$0
\$500Q	–	\$300Q	=	\$200,000	+	\$0
\$200Q = \$200,000						
$Q = \frac{\$200,000}{\$200} = \frac{\text{Fixed Costs}}{\text{Contribution Margin per Unit}}$						
Q = 1,000 units						
where						
Q = sales volume in units						
\$500 = selling price						
\$300 = variable costs per unit						
\$200,000 = total fixed costs						

Illustration 22-20

Contribution Margin Technique

- ◆ At the break-even point, contribution margin must equal total fixed costs

$$(\text{CM} = \text{total revenues} - \text{variable costs})$$

- ◆ Break-even point can be computed using either contribution margin per unit or contribution margin ratio.

Contribution Margin Technique

CONTRIBUTION MARGIN IN UNITS

- ◆ When the break-even-point in **units** is desired, contribution margin per unit is used in the following formula which shows the computation for Vargo Video:

Fixed Costs	÷	Contribution Margin per Unit	=	Break-Even Point in Units
\$200,000	÷	\$200	=	1,000 units

Illustration 22-21

Formula for break-even point
in units using unit contribution
margin

Contribution Margin Technique

CONTRIBUTION MARGIN RATIO

- ◆ When the break-even-point **in dollars** is desired, contribution margin ratio is used in the following formula which shows the computation for Vargo Video:

Fixed Costs	÷	Contribution Margin Ratio	=	Break-Even Point in Dollars
\$200,000	÷	40%	=	\$500,000

Illustration 22-22

Formula for break-even point
in dollars using contribution
Margin ratio

Graphic Presentation

Because this graph also shows costs, volume, and profits, it is referred to as a **cost-volume-profit (CVP) graph**.

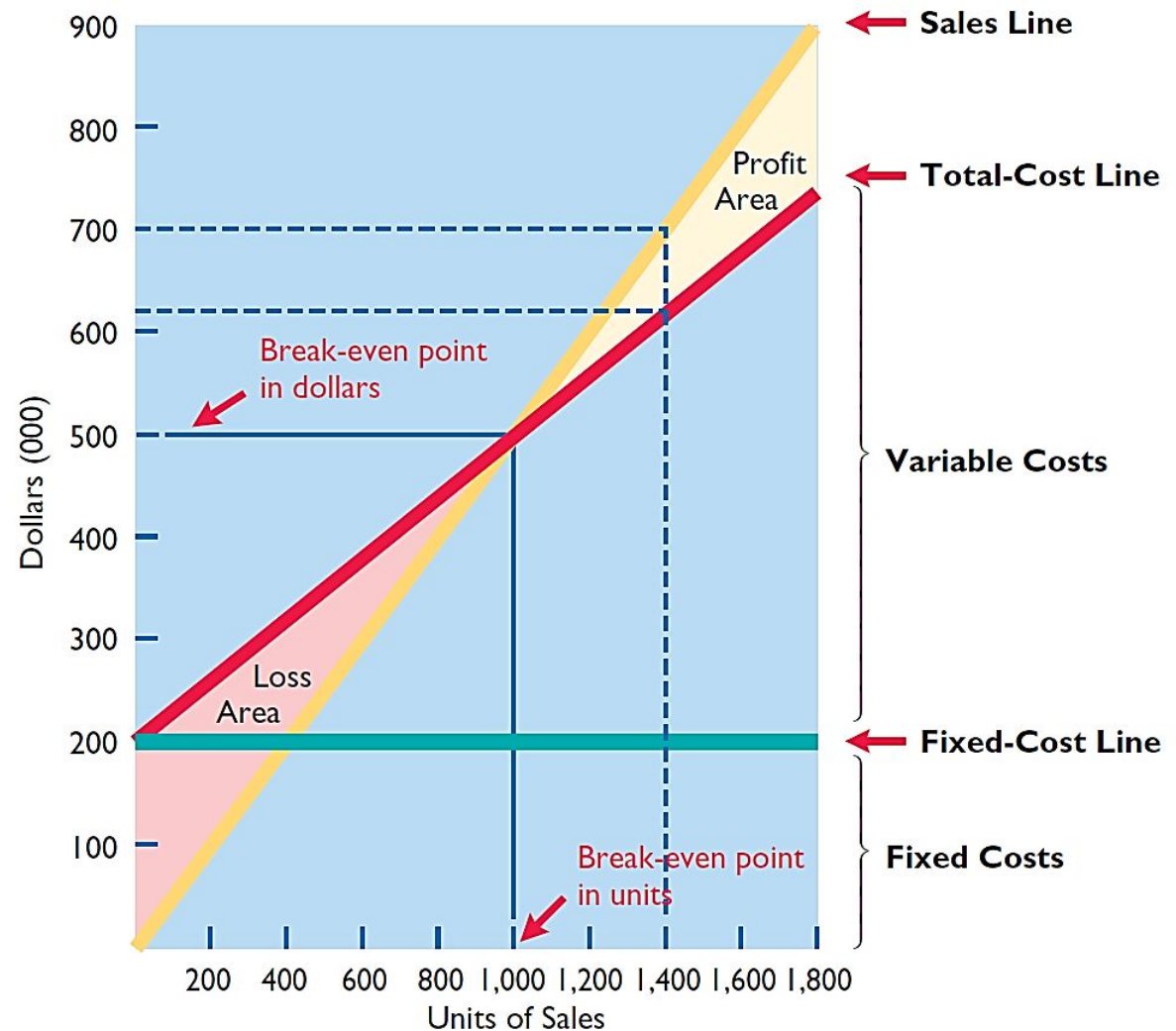
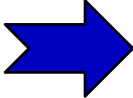


Illustration 22-23
CVP graph

Break-Even Analysis

Question

Gossen Company is planning to sell 200,000 pliers for \$4 per unit. The contribution margin ratio is 25%. If Gossen will break even at this level of sales, what are the fixed costs?

- a. \$100,000.
- b. \$160,000.
-  c. \$200,000.
- d. \$300,000.

DO IT!

4

Break-Even Analysis

Lombardi Company has a unit selling price of \$400, variable costs per unit of \$240, and fixed costs of \$180,000. Compute the **break-even point** in units using **(a) a mathematical equation** and (b) contribution margin per unit.



Sales	-	Variable Costs	-	Fixed Costs	=	Net Income
	-		-		=	
	-					
	=					

DO IT!

4

Break-Even Analysis

Lombardi Company has a unit selling price of \$400, variable costs per unit of \$240, and fixed costs of \$180,000. Compute the **break-even point** in units using (a) a mathematical equation and **(b) contribution margin per unit**.

$$\begin{array}{ccccc} \text{Fixed} & & \text{Contribution} & & \text{Break-Even} \\ \text{Costs} & \div & \text{Margin per Unit} & = & \text{Point in Units} \\ & & & & \\ & \div & & = & \end{array}$$

Target Net Income

- ◆ Level of sales necessary to achieve a specified income.
- ◆ Can be determined from each of the approaches used to determine break-even sales/units:
 - ▶ from a **mathematical equation**,
 - ▶ by using **contribution margin technique**, or
 - ▶ from a cost-volume profit (**CVP**) graph.
- ◆ Expressed either in **sales units** or in **sales dollars**.

Target Net Income

MATHEMATICAL EQUATION

Formula for required sales to meet target net income.

Illustration 22-24

Required Sales	–	Variable Costs	–	Fixed Costs	=	Target Net Income
-------------------	---	-------------------	---	----------------	---	----------------------

Target Net Income

MATHEMATICAL EQUATION

Using the formula for the break-even point, simply include the desired net income as a factor.

Illustration 22-25

Required Sales	–	Variable Costs	–	Fixed Costs	=	Target Net Income
\$500Q	–	\$300Q	–	\$200,000	=	\$120,000
\$500Q	–	\$300Q	=	\$200,000	+	\$120,000
$\$200Q = \$200,000 + \$120,000$						
$Q = \frac{\$200,000 + \$120,000}{\$200} = \frac{\text{Fixed Costs} + \text{Net Target Income}}{\text{Contribution Margin per Unit}}$						
$Q = 1,600$						
where						
Q = sales volume						
\$500 = selling price						
\$300 = variable costs per unit						
\$200,000 = total fixed costs						
\$120,000 = target net income						

Target Net Income

CONTRIBUTION MARGIN TECHNIQUE

To determine the required **sales in units** for Vargo Video:

$\left(\begin{array}{c} \text{Fixed Costs +} \\ \text{Target Net Income} \end{array} \right)$	$\div$	$\begin{array}{c} \text{Contribution} \\ \text{Margin per Unit} \end{array}$	$=$	Required Sales in Units
$(\$200,000 + \$120,000)$	$\div$	$\$200$	$=$	1,600 units

Illustration 22-26

Formula for required sales in units using unit contribution margin

Target Net Income

CONTRIBUTION MARGIN TECHNIQUE

To determine the required **sales in dollars** for Vargo Video:

$\left(\begin{array}{c} \text{Fixed Costs +} \\ \text{Target Net Income} \end{array} \right)$	$\div$	Contribution Margin Ratio	=	Required Sales in Dollars
$(\$200,000 + \$120,000)$	$\div$	40%	=	\$800,000

Illustration 22-27

Formula for required sales
in dollars using contribution
margin ratio

Target Net Income

GRAPHIC PRESENTATION

Suppose Vargo Video sells 1,400 camcorders. Illustration 22-23 shows that a vertical line drawn at 1,400 units intersects the sales line at \$700,000 and the total cost line at \$620,000. The difference between the two amounts represents the net income (profit) of \$80,000.

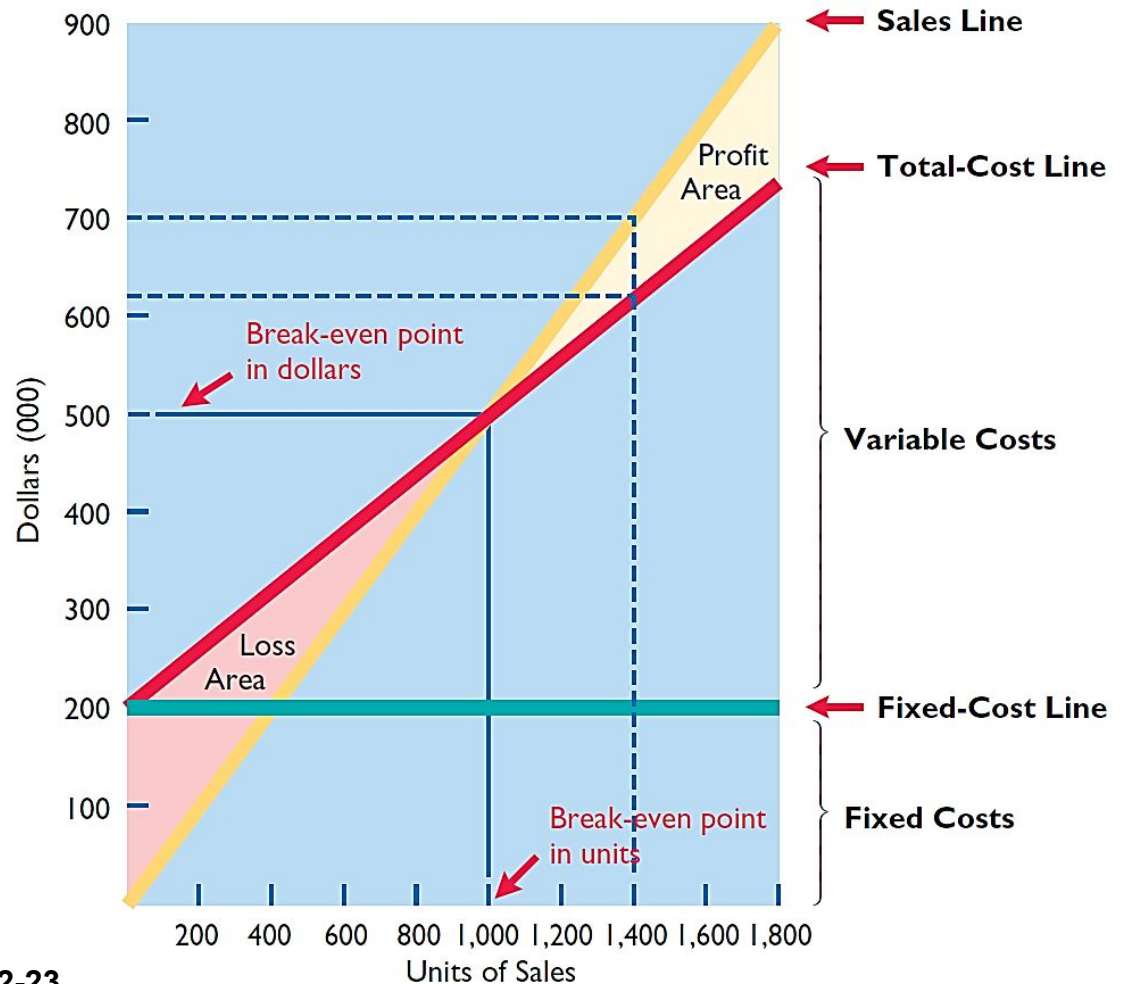


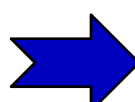
Illustration 22-23

Target Net Income

Question

The mathematical equation for computing required sales to obtain target net income is:

Required sales =

- a. Variable costs + Target net income.
-  b. Variable costs + Fixed costs + Target net income.
- c. Fixed costs + Target net income.
- d. No correct answer is given.

Margin of Safety

- ◆ Difference between **actual or expected sales** and sales at the **break-even point**.
- ◆ Measures the “cushion” that a particular level of sales provides.
- ◆ May be expressed **in dollars or as a ratio**.
- ◆ Assuming actual/expected sales are \$750,000:

Actual (Expected) Sales	—	Break-Even Sales	=	Margin of Safety in Dollars
\$750,000	—	\$500,000	=	\$250,000

Illustration 22-28

Formula for margin of safety
in dollars

Margin of Safety

- ◆ Computed by dividing the margin of safety in dollars by the actual (or expected) sales.
- ◆ Assuming actual/expected sales are \$750,000:

Illustration 22-29

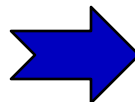
Actual (Expected) Sales	—	Break-Even Sales	=	Margin of Safety in Dollars
\$750,000	—	\$500,000	=	\$250,000

- ◆ The higher the dollars or percentage, the greater the margin of safety.

Margin of Safety

Question

Marshall Company had actual sales of \$600,000 when break-even sales were \$420,000. What is the margin of safety ratio?

- 
- a. 25%.
 - b. 30%.
 - c. 33 1/3%.
 - d. 45%.

DO IT!

5

Break-Even, Margin of Safety, and Target Net Income

Zootsuit Inc. makes travel bags that sell for \$56 each. For the coming year, management expects fixed costs to total \$320,000 and variable costs to be \$42 per unit. Compute the following:

- a) break-even point in dollars using the contribution margin (CM) ratio;
- b) the margin of safety and margin of safety ratio assuming actual sales are \$1,382,400; and
- c) the sales dollars required to earn net income of \$410,000.

DO IT!

5

Break-Even, Margin of Safety, and Target Net Income

Zootsuit Inc. makes travel bags that sell for \$56 each. For the coming year, management expects fixed costs to total \$320,000 and variable costs to be \$42 per unit. **Compute break-even point in dollars using the contribution margin (CM) ratio.**

Contribution margin ratio =

Break-even sales in dollars =

DO IT!

5

Break-Even, Margin of Safety, and Target Net Income

Zootsuit Inc. makes travel bags that sell for \$56 each. For the coming year, management expects fixed costs to total \$320,000 and variable costs to be \$42 per unit. **Compute the margin of safety and margin of safety ratio assuming actual sales are \$1,382,400.**

Margin of safety =

Margin of safety ratio =

DO IT!

5

Break-Even, Margin of Safety, and Target Net Income

Zootsuit Inc. makes travel bags that sell for \$56 each. For the coming year, management expects fixed costs to total \$320,000 and variable costs to be \$42 per unit. **Compute the sales dollars required to earn net income of \$410,000.**

Required sales in dollars =

Use CVP analysis to respond to changes in the business environment.

Illustration: Three independent situations that might occur at Vargo Video. Each case uses the original camcorder sales and cost data, which were as follows.

Unit selling price	\$500
Unit variable cost	\$300
Total fixed costs	\$200,000
Break-even sales	\$500,000 or 1,000 units

Illustration 22-30

Original camcorder sales
and cost data

Case I: Offering a Discount

A competitor is offering a 10% discount on the selling price of its camcorders. Management must decide whether to offer a similar discount.

Question: What effect will a 10% discount on selling price (\$500 x 10% = \$50) have on the breakeven point?

Fixed Costs	÷	Unit Contribution Margin	=	Break-Even Sales
\$200,000	÷	\$150	=	1,333 units (rounded)

Illustration 22-31
Computation of break-even
sales in units

Case II: Investing in New Equipment

Management invests in new robotic equipment that will lower the amount of direct labor required to make camcorders. Estimates are that total fixed costs will increase 30% and that variable cost per unit will decrease 30%.

Question: What effect will the new equipment have on the sales volume required to break even?

Fixed Costs	÷	Unit Contribution Margin	=	Break-Even Sales
\$260,000	÷	(\$500 - \$210)	=	897 units (rounded)

Illustration 22-32
Computation of break-even
sales in units

Case III: Determining Required Sales

Vargo's principal supplier of raw materials has just announced a price increase. The higher cost is expected to increase the variable cost of camcorders by \$25 per unit. Management decides to hold the line on the selling price of the camcorders. It plans a cost-cutting program that will save \$17,500 in fixed costs per month. Vargo is currently realizing monthly net income of \$80,000 on sales of 1,400 camcorders.

Question: What increase in units sold will be needed to maintain the same level of net income?

Case III: Determining Required Sales

Variable cost per unit increases to _____

Fixed costs are reduced to _____

Contribution margin per unit becomes _____

(Fixed Cost + Target Net Income)	-	Unit Contribution Margin	=	Required Sales in Units
(\$182,500 + \$80,000)	-	\$175	=	1,500

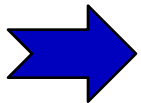
Illustration 22-33
Computation of required sales

Basic Concepts

Question

Croc Catchers calculates its contribution margin to be less than zero. Which statement is true?

- a. Its fixed costs are less than the variable cost per unit.
- b. Its profits are greater than its total costs.
- c. The company should sell more units.
- d. Its selling price is less than its variable costs.



Don't Just Look—Buy Something

When analyzing an Internet business such as **Amazon. com**, analysts closely watch the so-called “conversion rate.” This rate is calculated by dividing the number of people who actually take action at an Internet site (buy something) by the total number of people who visit the site. Average conversion rates are from 3% to 5%. A rate below 2% is poor, while a rate above 10% is great. Conversion rates have an obvious effect on the breakeven point. Suppose you spend \$10,000 on your site, which then attracts 5,000 visitors. If you get a 2% conversion rate (100 purchases), your site costs \$100 per purchase ($\$10,000 \div 100$). A 4% conversion rate lowers your cost to \$50 per transaction, and an 8% conversion rate gets you down to \$25. Studies show that conversion rates increase if the site has an easy-to-use interface, fast-performing screens, a convenient ordering process, and advertising that is both clever and clear.

Sources: J. William Gurley, “The One Internet Metric That Really Counts” *Fortune* (March 6, 2000), p. 392; and Milind Mody, “Chief Mentor: How Startups Can Win Customers Online,” *Wall Street Journal Online*, (May 11, 2011).

CVP Income Statement Revisited

Assume that Vargo Video reaches its target net income of \$120,000. The following information is obtained on the \$680,000 of costs that were incurred in June to produce and sell 1,600 units.

	<u>Variable</u>	<u>Fixed</u>	<u>Total</u>
Cost of goods sold	\$400,000	\$120,000	\$520,000
Selling expenses	60,000	40,000	100,000
Administrative expenses	20,000	40,000	60,000
	<u>\$480,000</u>	<u>\$200,000</u>	<u>\$680,000</u>

Illustration 22-34
Assumed cost and
expense data

VARGO VIDEO COMPANY
CVP Income Statement
For the Month Ended June 30, 2017

	<u>Total</u>	<u>Per Unit</u>
Sales	\$ 800,000	\$ 500
Variable expenses		
Cost of goods sold	\$400,000	
Selling expenses	60,000	
Administrative expenses	<u>20,000</u>	
Total variable expenses	<u>480,000</u>	<u>300</u>
Contribution margin	320,000	<u>\$200</u>
Fixed expenses		
Cost of goods sold	120,000	
Selling expenses	40,000	
Administrative expenses	<u>40,000</u>	
Total fixed expenses	<u>200,000</u>	
Net income	<u>\$120,000</u>	

Illustration 22-35

Detailed CVP income statement

DO IT!**6**

CVP Analysis

Krisanne Company reports the following operating results for the month of June 2017.

	<u>Total</u>	<u>Per Unit</u>
Sales (5,000 units)	\$300,000	\$60
Variable costs	<u>180,000</u>	<u>36</u>
Contribution margin	120,000	<u>\$24</u>
Fixed expenses	<u>100,000</u>	
Net income	<u><u>\$ 20,000</u></u>	

To increase net income, management is considering reducing the selling price by 10%, with no changes to unit variable costs or fixed costs. Management is confident that this change will increase unit sales by 25%. Using the contribution margin technique, compute the break-even point in units and dollars and margin of safety in dollars (a) assuming no changes to sales price or costs, and (b) assuming changes to sales price and volume as described above. (c) Comment on your findings.

DO IT!**6**

CVP Analysis

Krisanne Company reports the following operating results for the month of June 2017.

	<u>Total</u>	<u>Per Unit</u>
Sales (5,000 units)	\$300,000	\$60
Variable costs	<u>180,000</u>	<u>36</u>
Contribution margin	120,000	<u>\$24</u>
Fixed expenses	<u>100,000</u>	
Net income	<u><u>\$ 20,000</u></u>	

Solution

(a) Assuming no changes to sales price or costs:

Break-even point in units =

Break-even point in sales dollars =

Margin of safety in dollars =

DO IT!**6**

CVP Analysis

Krisanne Company reports the following operating results for the month of June 2017.

	<u>Total</u>	<u>Per Unit</u>
Sales (5,000 units)	\$300,000	\$60
Variable costs	<u>180,000</u>	<u>36</u>
Contribution margin	120,000	<u>\$24</u>
Fixed expenses	<u>100,000</u>	
Net income	<u>\$ 20,000</u>	

Solution

(b) Assuming changes to sales price and volume:

Break-even point in units =

Break-even point in sales dollars =

Margin of safety in dollars =

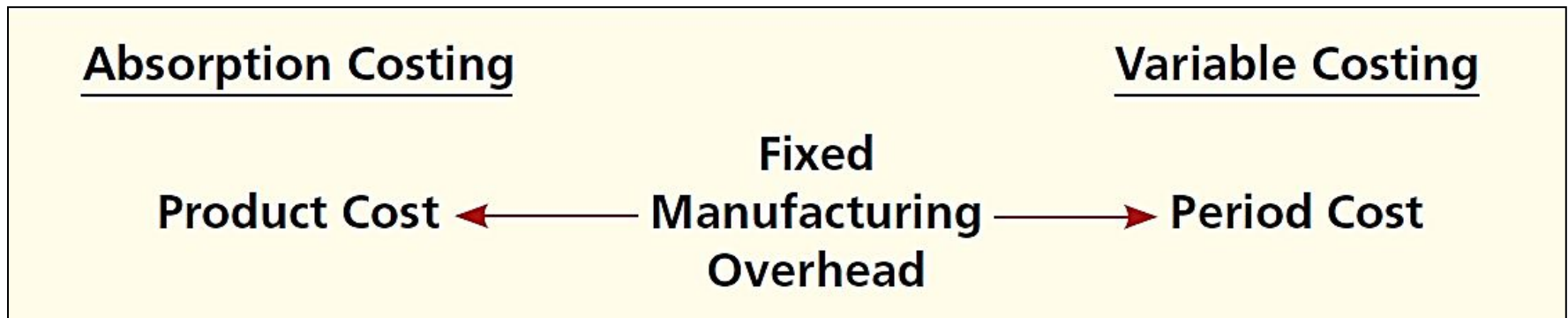
Krisanne Company reports the following operating results for the month of June 2017.

	<u>Total</u>	<u>Per Unit</u>
Sales (5,000 units)	\$300,000	\$60
Variable costs	<u>180,000</u>	<u>36</u>
Contribution margin	120,000	<u>\$24</u>
Fixed expenses	<u>100,000</u>	
Net income	<u><u>\$ 20,000</u></u>	

- (c) The increase in the break-even point and the decrease in the margin of safety indicate that management should not implement the proposed change. The increase in sales volume will result in contribution margin of \$112,500 ($6,250 \times \18), which is \$7,500 less than the current amount.

APPENDIX 22A: Explain the difference between absorption costing and variable costing.

Under **variable costing** only direct materials, direct labor, and variable manufacturing overhead costs are considered product costs. Companies recognize fixed manufacturing overhead costs as period costs (expenses) when incurred.

**Illustration 22A-1**

Difference between absorption costing and variable costing

CVP Income Statement Revisited

Illustration: Assume that Premium Products Corporation manufactures a polyurethane sealant, called Fix-It, for car windshields. Relevant data for Fix-It in January 2017, the first month of production, are as follows.

Selling price	\$20 per unit.
Units	Produced 30,000; sold 20,000; beginning inventory zero.
Variable unit costs	Manufacturing \$9 (direct materials \$5, direct labor \$3, and variable overhead \$1).
	Selling and administrative expenses \$2.
Fixed costs	Manufacturing overhead \$120,000.
	Selling and administrative expenses \$15,000.

Illustration 22A-2
Sealant sales and cost data for
Premium Products Corporation

CVP Income Statement Revisited

Illustration: The per unit production cost of Fix-It under each costing approach is:

Illustration 22A-3
Computation of per unit manufacturing cost

Type of Cost	Absorption Costing	Variable Costing
Direct materials	\$ 5	\$5
Direct labor	3	3
Variable manufacturing overhead	1	1
Fixed manufacturing overhead (\$120,000 ÷ 30,000 units produced)	4	0
Manufacturing cost per unit	<u>\$13</u>	<u>\$9</u>

Based on these data, each unit sold and each unit remaining in inventory is costed under absorption costing at \$13 and under variable costing at \$9.

PREMIUM PRODUCTS CORPORATION

Income Statement For the Month Ended January 31, 2017 Absorption Costing

Sales (20,000 units $\times$ \$20)		\$400,000
Cost of goods sold		
Inventory, January 1	\$ -0-	
Cost of goods manufactured (30,000 units $\times$ \$13)	<u>390,000</u>	
Cost of goods available for sale	390,000	
Less: Inventory, January 31 (10,000 units $\times$ \$13)	<u>130,000</u>	
Cost of goods sold (20,000 units $\times$ \$13)		<u>260,000</u>
Gross profit		140,000
Variable selling and administrative expenses (20,000 $\times$ \$2)	40,000	
Fixed selling and administrative expenses	<u>15,000</u>	<u>55,000</u>
Net income		<u><u>\$ 85,000</u></u>

Illustration 22A-4
Absorption costing
income statement

Helpful Hint

The income statement format in Illustration 22A-4 is the same as that used under generally accepted accounting principles.

PREMIUM PRODUCTS CORPORATION
Income Statement
For the Month Ended January 31, 2017
Variable Costing

Sales (20,000 units × \$20)		\$400,000
Variable cost of goods sold		
Inventory, January 1	\$ -0-	
Variable cost of goods manufactured (30,000 units × \$9)	<u>270,000</u>	
Variable cost of goods available for sale	270,000	
Less: Inventory, January 31 (10,000 units × \$9)	<u>90,000</u>	
Variable cost of goods sold	180,000	
Variable selling and administrative expenses (20,000 units × \$2)	<u>40,000</u>	<u>220,000</u>
Contribution margin		180,000
Fixed manufacturing overhead	120,000	
Fixed selling and administrative expenses	<u>15,000</u>	<u>135,000</u>
Net income		<u><u>\$ 45,000</u></u>

Illustration 22A-5
Variable costing
income statement

Helpful Hint

Note the difference in the computation of the ending inventory:
\$9 per unit here, \$13 per unit in Illustration 22A-4.

Absorption and Variable Costing

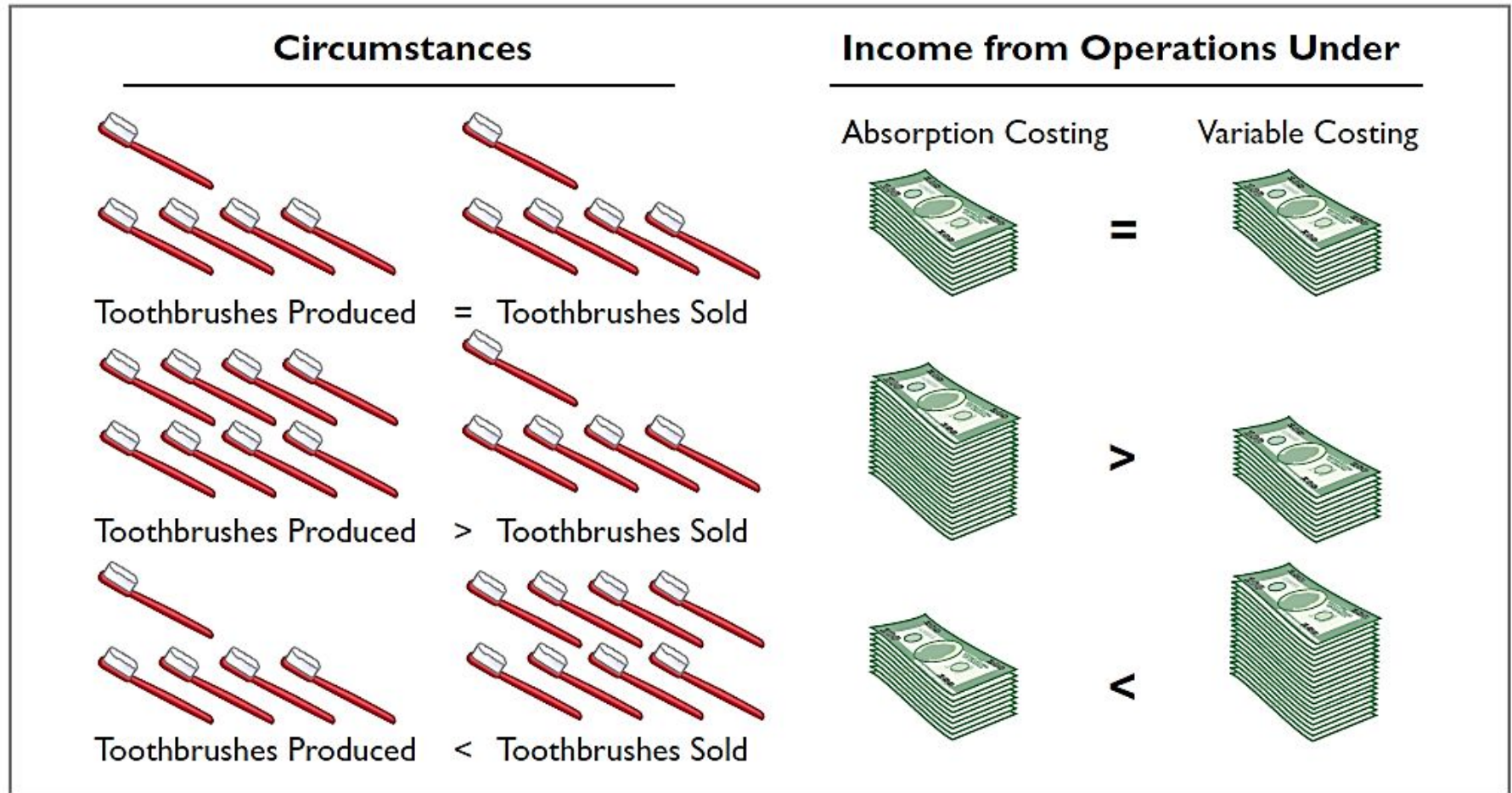


Illustration 22A-6

Summary of income effects under absorption costing and variable costing

Rationale for Variable Costing

- ◆ The purpose of fixed manufacturing costs is **to have productive facilities available for use.**
- ◆ The use of variable costing is acceptable **only for internal use by management.**

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